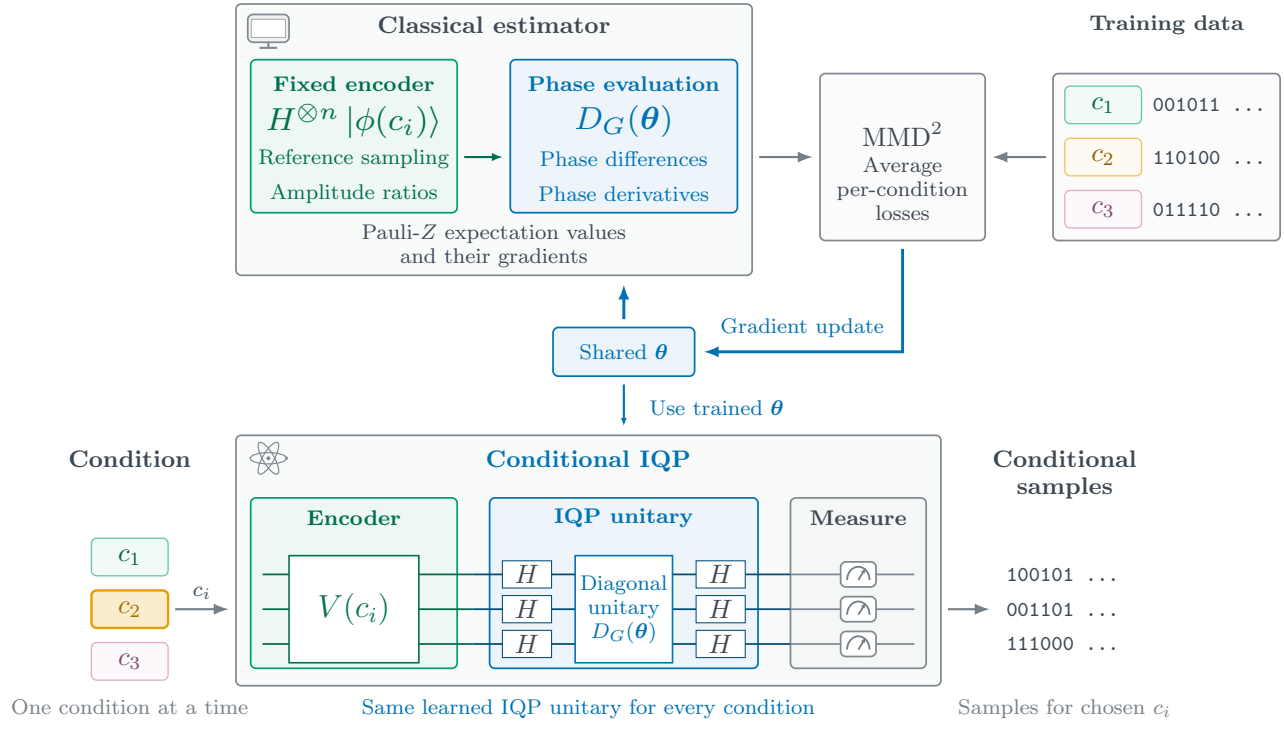

Classical estimator
Fixed encoder
$H^{\otimes n}\,|\phi(c_i)\rangle$
Reference sampling
Amplitude ratios
Phase evaluation
$D_G(\boldsymbol{\theta})$
Phase differences
Phase derivatives
Pauli-Z expectation values
and their gradients
Training data
c_1 001011 ...
c_2 110100 ...
c_3 011110 ...
MMD^2
Average
per-condition
losses
Gradient update
Shared $\boldsymbol{\theta}$
Use trained $\boldsymbol{\theta}$
Condition
c_1
c_2
c_3
c_i
Conditional IQP
Encoder
$V(c_i)$
IQP unitary
H
Diagonal
unitary
$D_G(\boldsymbol{\theta})$
Measure
Conditional
samples
100101 ...
001101 ...
111000 ...
One condition at a time
Same learned IQP unitary for every condition
Samples for chosen c_i